4.2 Non-SAE Baselines

In order to compare our setup to previous work [Heimersheim and Mendel, 2024], we run perturbations towards model-generated (real) and random activations. We sample 1000 prompts from the dataset and run inference on them to obtain the base activations, and perturb each base activation in two directions:

1. **Model-generated (real):** Towards a randomly selected activation produced by the model.
2. **Random:** Towards a randomly sampled point from a normal distribution with the same mean and covariance as model-generated activations (calculated using $32,000$ model-generated activations).

We plot examples of perturbations towards real and random activations in Figure 1. The distribution of distances between the base activation and the target are similar for both baselines, with a mean of $\simeq 40$ in activation space. Additionally, we find that the average cosine similarity between two model-generated activations with respect to the SAE decoder bias is $\simeq 0.42$.

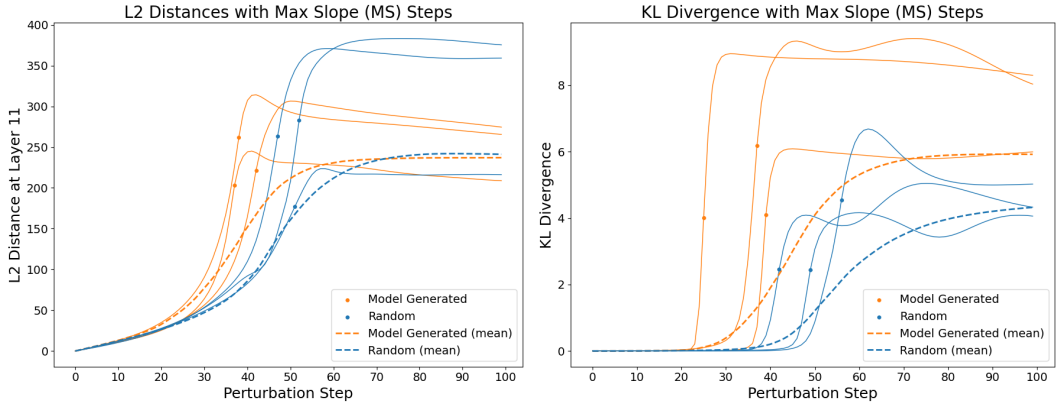

Figure 1: The L2 distance after Layer 11 (left) and the KL divergence of the next-token prediction probabilities (right) between the perturbed and unperturbed model, as three base activations at Layer 1 are slowly perturbed towards model-generated activations (orange) and random points sampled from the distribution of model activations (blue). The x-axis represents the total length of the perturbation broken into 100 steps of size 0.5 each. The dot on each solid line represents the maximum slope (MS) step for each perturbation. The dashed lines represent the average L2 distance and KL divergence per step for 1000 perturbations of both types. The linear part at the start of the curves represents the activation plateau, and the sharp rise in the curves represents the blowup.

4.3 Synthetic Activations

We construct synthetic activations using three methods that each use different amounts of information about SAE latents, and compare which ones match the behavior of real activations.

Our simplest approach (**synthetic-random**) randomly selects SAE latents and assigns them the same latent activations as the active latents of the base activation. We consider this a weak approach to compose synthetic activations and mainly focus on the next two methods, but we show the results for this method in Appendix A.

Our second approach (**synthetic-baseline**) accounts for SAE latent sparsities along with latent activations, matching the distribution of latent sparsities from the base activation:

1. We use the SAE to encode the base activation and obtain its active latents and their latent activations.
2. We replace each active SAE latent with a new one sampled from 10 most similarly-sparse latents and assign it the latent activation of the original active latent.
3. We decode the new latent activations to obtain the **synthetic-baseline** activation.

For our third and final method (**synthetic-structured**) we capture and reproduce geometric properties that SAE latents of real activations have. We get the best results when we control for sparsity of latents and the cosine similarity relationship between latents. The procedure for the generation of synthetic-structured activations is as follows:

1. We use the SAE to encode the base activation and obtain its active latents and their latent activations. The active latent with the highest latent activation is the top latent of the base activation (`top_base`).
2. We create a list of 100 non-dead SAE latents with the most similar sparsity to `top_base`.
3. Out of the 100 selected latents, we select one latent that has cosine similarity closest to 0.42 (mean cosine similarity between two real activations w.r.t. the SAE decoder bias) with `top_base`.
4. This latent becomes the top latent for our synthetic activation (`top_synth`), and we give it a latent activation value equal to that of `top_base`.
5. For each remaining active latent (`l_base`) in the base activation:
   (a) We calculate its cosine similarity (`l_top_cos_sim`) with `top_base`.
   (b) We select a latent (`l_synth`) that has cosine similarity with `top_synth` equal to `l_top_cos_sim`.
   (c) We assign `l_synth` a latent activation value equal to that of `l_base`.
6. We construct a latent activation vector with zeros for all latents except the latents selected above, and decode it to obtain the **synthetic-structured** activation.

We perform 1000 perturbations towards synthetic-baseline and synthetic-structured activations each, using the setup described in Section 4.1, where the direction D is the normalised difference between the synthetic activation and the base activation.

4.4 Activation Plateaus

If our synthetic activations behave like real activations, they should not only reproduce directional sensitivity behavior, but also exhibit activation plateaus. To test this, we use the following perturbation approach:

1. We initiate perturbations from four distinct base activations: model-generated activations, synthetic activations generated using SAE latents of similar sparsity to the base activation (synthetic-baseline), synthetic activations generated using selected SAE latents based on sparsity and cosine similarity relationship (synthetic-structured), and random points sampled from the distribution described in Section 4.2 (random).
2. For all four starting points, we perturb towards a random activation.
3. We track how quickly L2 distance at Layer 11 increases near the start of the perturbation by recording the activation plateau perturbation step (AP step) at which the L2 distance between the unperturbed and perturbed models crosses a value of 20 (we found that a simple threshold was enough to distinguish the behavior of different activations).

The AP step measures the flatness of the activation plateau around activations, with larger AP steps signify flatter activation plateaus. We repeat this process for 1000 perturbations and collect the distributions of AP steps for each activation type.

5 Results

We find that the behavior of synthetic activations as compared to random activations and real activations differs for the two experiments we perform. This suggests that directional sensitivity and activation plateaus point to different properties of SAE latents that make up real activations (see Appendix E for details on properties that we account for). We now provide detailed results for each of our experiments below.

5.1 Directional Sensitivity

For this experiment, we perturb real activations towards different types of activations we construct and study the model's sensitivity to these perturbations. Figure 2 shows the distributions of max slope (MS) steps for L2 distance for perturbations towards synthetic-baseline and synthetic-structured activations compared to model-generated (real) and random activations. We also provide statistics for the MS step distributions for perturbations towards all activations types in Table 1.

When comparing the MS steps (Table 1) for perturbations towards real and random activations, we find that perturbations towards real activations cause earlier (lower mean) and more localized (lower variance) blowups than perturbations towards random activations. This means that perturbing towards real activations affects the model output more than perturbing towards random activations.

While synthetic-baseline activations do not fully replicate the behavior of real activations, they still perform better than random activations (Figure 2). We calculate similarity between two MS step distributions using the Kolmogorov Smirnov (KS) statistic [Smirnov, 1948]. This demonstrates that the model is more sensitive to perturbations towards synthetic activations composed of SAE latents than perturbations towards randomly sampled points from the distribution of model activations (random). This suggests that SAE latents encode more information about model computation than random directions do.

Importantly, perturbations towards synthetic-structured activations look a lot more similar to perturbations towards model-generated activations than perturbations towards synthetic-baseline and random activations do (Figure 2, Table 1). This implies that relationships between SAE latents are important, and that model-generated activations are not approximated well by "bags of SAE latents". Additionally, we find that synthetic-random activations perform worse than synthetic-baseline activations, confirming our decision to use the latter as a stronger baseline (see Appendix A for more details on synthetic-random activations).

The distance between the base activation and the target activation varies for each perturbation we perform, and we find that this can directly influence the location of the blowup in our setup. In order to remove the effect of the distance on the MS step distribution, we also perform perturbations with relative step size (see Appendix A for details). The gap between the synthetic-structured and synthetic-baseline reduces in the relative step size setup. This is because synthetic-baseline activations are further away from base activations than synthetic-structured activations, and hence cause blowups later.

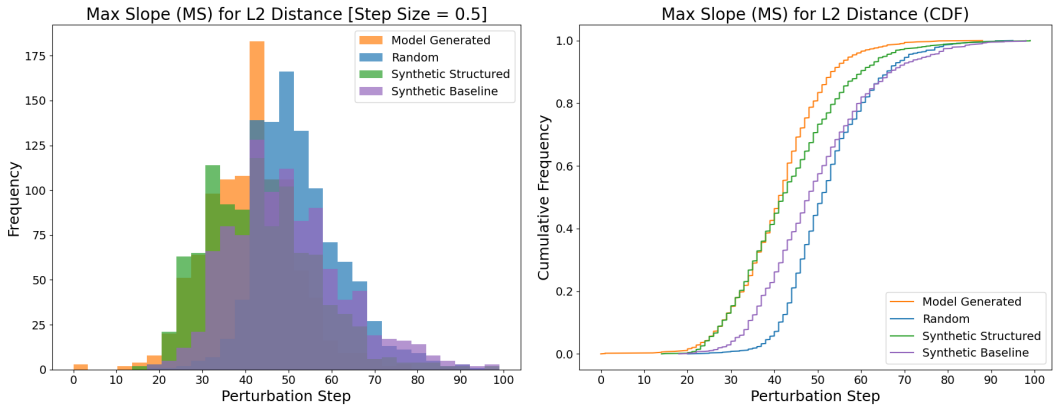

Figure 2: The distributions of the max slope (MS) steps for perturbations towards model-generated (orange), random (blue), synthetic-baseline (purple), and synthetic-structured (green) activations. The left panel shows the counts of MS steps occurring in different bins along the length of the perturbation, and the right panel shows corresponding cumulative frequency. We find that perturbing towards synthetic-structured activations is more similar to perturbing towards model-generated activations as compared to perturbing towards synthetic-baseline activations.